

Kairos — Test Protocol

Project: Kairos Time Tracking & Invoicing **Date:** 2026-03-28 **Prepared by:** Claude Code (Anthropic) **Test Frameworks:** Vitest (unit tests) · Playwright (UAT / E2E)

1. Scope

This protocol documents all automated tests written for the Kairos application. Tests cover three layers:

Layer	Framework	Location	Count
Unit — Business Logic	Vitest	__tests__/unit/	67 tests
UAT — User Acceptance	Playwright	__tests__/e2e/	40 tests

2. Unit Tests

Unit tests validate pure business logic functions in isolation — no browser, no database, no network calls.

2.1 Utility Functions — `formatDuration` , `formatMoney` , `calcEntryEarnings`

File: `__tests__/unit/lib/types.test.ts`

#	Test Case	Expected Result
U-01	<code>formatDuration(0)</code>	"00:00:00"
U-02	<code>formatDuration(45)</code> — sub-minute	"00:00:45"
U-03	<code>formatDuration(60)</code> — exactly one minute	"00:01:00"
U-04	<code>formatDuration(90)</code> — minutes and seconds	"00:01:30"
U-05	<code>formatDuration(3600)</code> — exactly one hour	"01:00:00"
U-06	<code>formatDuration(3661)</code> — hours, minutes, seconds	"01:01:01"
U-07	<code>formatDuration(28800)</code> — 8 working hours	"08:00:00"
U-08	<code>formatDuration(3723)</code> — leading zero padding	"01:02:03"
U-09	<code>formatDuration(360000)</code> — over 100 hours	"100:00:00"
U-10	<code>formatMoney(0)</code> — zero EUR	Contains € and 0
U-11	<code>formatMoney(1000)</code> — integer amount	Contains €, 1, 000
U-12	<code>formatMoney(1234.56)</code> — decimals	Contains €, 234, 56
U-13	<code>formatMoney(-500)</code> — negative	Contains - and 500
U-14	<code>formatMoney(100)</code> — default currency EUR	Contains €
U-15	<code>formatMoney(100, 'USD')</code> — custom currency	Contains \$ or usd

U-16	<code>calcEntryEarnings(0, project, null)</code> — zero duration	Returns 0
U-17	<code>calcEntryEarnings(3600, null, null)</code> — null project	Returns 0
U-18	<code>calcEntryEarnings(3600, undefined, null)</code> — undefined project	Returns 0
U-19	1 hour at €100/h, no level	Returns 100
U-20	30 min at €120/h (fractional hours)	Returns 60
U-21	Level rate matches <code>levelId</code> — uses level rate €80 not project rate €100	Returns 80
U-22	<code>levelId</code> has no matching rate — falls back to project rate	Returns 100
U-23	Empty <code>level_rates</code> array — falls back to project rate	Returns 90
U-24	Multiple level rates — picks correct senior rate €150	Returns 150
U-25	Project with zero hourly rate	Returns 0
U-26	8 hours at €125/h	Returns 1000

2.2 Invoice Logic — Classification & Generation

File: `__tests__/unit/api/invoice-logic.test.ts`

classifyEntries — Hours Summary

#	Test Case	Expected Result
U-27	Entry on approved timesheet → approved hours	<code>approvedHours</code> = 1, others = 0
U-28	Entry on submitted timesheet → pending hours	<code>pendingHours</code> = 1
U-29	Entry with no timesheet → draft hours	<code>draftHours</code> = 1
U-30	Entry with draft timesheet → draft hours	<code>draftHours</code> = 1
U-31	Approved revenue uses per-entry <code>hourly_rate</code> (€150) not project rate	<code>approvedRevenue</code> = 150
U-32	Entry rate €80 overrides project rate €100	<code>approvedRevenue</code> = 80
U-33	Multiple entries same project → hours summed	<code>approvedHours</code> ≈ 1.5
U-34	Two users on same project — one approved, one submitted	<code>approvedHours</code> = 1, <code>pendingHours</code> = 2
U-35	Entries on different projects → separate summaries	Returns 2 distinct summaries

buildInvoiceLines — Invoice Generation

#	Test Case	Expected Result
U-36	Entries from 2 different projects	Returns 2 invoice lines
U-37	2 entries on same project, 1h each	hours = 2
U-38	1h at €80 + 1h at €120	amount = 200 (per-entry rates)
U-39	1h at €80 + 1h at €120	Blended rate = 100
U-40	Single entry	description = "Alpha" (project name)
U-41	1h 20m (4800 seconds)	hours = 1.33 (rounded to 2dp)
U-42	1h 20m at €100/h	amount = 133.33 (rounded to 2dp)
U-43	Empty input	Returns empty array

calcCashflow — Analytics KPI

#	Test Case	Expected Result
U-44	Paid invoice €1000	billed = 1000, paid = 1000
U-45	Sent invoice (not yet paid)	billed = 500, paid = 0
U-46	Sent invoice past due date	overdue = 300, open = 0
U-47	Sent invoice future due date	open = 400, overdue = 0
U-48	Draft invoice	All values = 0 (drafts ignored)
U-49	Mixed: paid €1000 + open €500 + overdue €200 + draft €9999	billed = 1700, paid = 1000, open = 500, overdue = 200
U-50	Empty invoice list	{ billed: 0, paid: 0, open: 0, overdue: 0 }

2.3 PM Timesheet Authorization

File: `__tests__/unit/api/timesheet-pm-auth.test.ts`

canPMReviewTimesheet — Authorization

#	Test Case	Expected Result
U-51	PM reviews timesheet of their own project member	Allowed (<code>true</code>)
U-52	PM reviews timesheet of a member on another PM's project	Denied (<code>false</code>)

U-53	PM manages no projects	Denied (<i>false</i>)
U-54	User is not a member of any of PM's projects	Denied (<i>false</i>)
U-55	Project is soft-deleted	Denied (<i>false</i>)
U-56	Cross-workspace check — member in ws-1, checking ws-2	Denied (<i>false</i>)
U-57	Member on multiple projects, one belongs to the PM	Allowed (<i>true</i>)
U-58	PM tries to review their own timesheet	Denied (<i>false</i>)

Timesheet Status Transitions

#	Test Case	Expected Result
U-59	Approve a submitted timesheet	Valid (<i>true</i>)
U-60	Reject a submitted timesheet	Valid (<i>true</i>)
U-61	Approve a draft timesheet	Invalid (<i>false</i>)
U-62	Re-approve an already approved timesheet	Invalid (<i>false</i>)
U-63	Submit a draft timesheet	Valid (<i>true</i>)
U-64	Re-submit a rejected timesheet	Valid (<i>true</i>)
U-65	Submit an already submitted timesheet	Invalid (<i>false</i>)
U-66	Withdraw a submitted timesheet	Valid (<i>true</i>)
U-67	Withdraw a draft timesheet	Invalid (<i>false</i>)

3. UAT / E2E Tests (Playwright)

All E2E tests run against a live instance of the application (`http://localhost:3000`) using a real Supabase database. Tests authenticate as the Partner account and reuse a stored session (`__tests__/e2e/.auth/session.json`).

Test account: `maximilian.stubhan@gmx.at` (Partner / Admin role)

3.1 Authentication

File: `__tests__/e2e/auth.setup.ts`

#	Test Case	Expected Result
E-01	Login with valid Partner credentials	Redirected to <code>/dashboard</code> , session stored

3.2 Dashboard

File: `__tests__/e2e/dashboard.spec.ts`

#	Test Case	Expected Result
E-02	Dashboard loads and shows utilization KPI	"Utilization" text visible
E-03	This Week and This Month KPI cards present	Both labels visible
E-04	Sidebar navigation links visible	Timer, Projects, Timesheets links present
E-05	Partner role badge is displayed	"Partner" badge visible

3.3 Timer

File: `__tests__/e2e/timer.spec.ts`

#	Test Case	Expected Result
E-06	Timer page loads with heading	"Time Tracker" or "Timer" heading visible
E-07	Start button is visible	Start button rendered
E-08	Clicking Start activates a running timer	Stop/square button appears within 5s
E-09	Clicking Stop creates a time entry	Entry row appears in list within 8s
E-10	Manual entry tab allows from/to time input	Start and End inputs visible (if tab exists)
E-11	Project dropdown lists available projects	Dropdown with at least 1 option
E-12	Entry list shows recent entries or empty state	Table or empty state visible

3.4 Timesheets

File: `__tests__/e2e/timesheets.spec.ts`

#	Test Case	Expected Result
E-13	Timesheets page loads with heading	"Timesheets" heading visible
E-14	"My Timesheets" and "Review" tabs present (Partner)	Both tab buttons visible
E-15	Review tab is active by default for Partner	Review tab has active styling
E-16	My Timesheets tab shows week navigation (KW XX)	Calendar week number visible
E-17	Week navigation prev/next chevrons present	At least 1 chevron button
E-18	Shows tracked hours for current week	Hours in format x.xh visible
E-19	Submit button disabled when 0 hours tracked	Submit button disabled (if 0h week)
E-20	Review tab shows team timesheets or empty state	Card list or "no timesheets" message

3.5 Invoices

File: `__tests__/e2e/invoices.spec.ts`

#	Test Case	Expected Result
E-21	Invoices page loads with heading	"Invoices" heading visible
E-22	Generate Invoice and Saved Invoices tabs present	Both tab buttons visible
E-23	Generate tab shows client selector	Client dropdown visible
E-24	Generate tab shows date range pickers	From Date and To Date inputs visible
E-25	Generate Invoice button disabled until client selected	Button not enabled initially
E-26	Selecting a client loads hours summary panel	"Hours available to bill" text appears (if clients exist)
E-27	Generate button disabled when no approved hours	Button disabled (if 0 approved hours)
E-28	History tab shows saved invoices or empty state	Cards or "no saved invoices" message
E-29	History invoice cards show amount (€) and status badge	€-amount and sent/paid/draft badge visible (if invoices exist)

3.6 Analytics

File: `__tests__/e2e/analytics.spec.ts`

#	Test Case	Expected Result
E-30	Analytics page loads with heading	"Analytics" heading visible
E-31	4 core KPI cards visible	Revenue MTD, Pipeline, Utilization, Effective Rate
E-32	Cashflow KPI section visible	"Cashflow" heading, Total Billed, Collected, Outstanding, Overdue
E-33	Team Utilization section visible	"Team Utilization" label
E-34	Utilization range selector has expected options	This week, This month, Last month, Custom range
E-35	Clicking a team member drills into weekly breakdown	"All Members" back button appears
E-36	Custom range inputs appear when "Custom" is selected	Two date inputs visible

E-37	Revenue trend chart rendered as SVG	SVG element visible
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3.7 Projects

File: `__tests__/e2e/projects.spec.ts`

#	Test Case	Expected Result
E-38	Projects page loads with heading	"Projects" heading visible
E-39	Shows project list or empty state	Cards or "no projects" message
E-40	"New Project" button visible for Partner	Button rendered
E-41	Clicking New Project opens a modal/form	Dialog or form with project fields appears
E-42	Project form has a Name field	Name input visible
E-43	Project form shows Project Manager dropdown	"Project Manager" label visible
E-44	Existing project cards show project info	Cards with €/hours data visible (if projects exist)
E-45	Project card shows Crown icon when PM is assigned	Crown SVG icon visible (if PM assigned)

4. Bugs Fixed During This Test Cycle

The following bugs were discovered and resolved prior to test implementation:

ID	Module	Bug Description	Fix
BUG-01	Timesheets API	PM could not review timesheets — 403 Forbidden	Authorization now checks <code>project_members</code> table instead of <code>time_entries</code>
BUG-02	Invoices	Invoice amount showed €0 after creation	<code>generate()</code> now uses per-entry <code>hourly_rate</code> snapshot; blended rate calculated correctly
BUG-03	Invoices	Save & Send button did nothing (silent fail)	Missing <code>created_by</code> (NOT NULL) added to insert payload; errors now surfaced
BUG-04	Timesheets UI	PM could not see team members' timesheets	<code>pmUserIds</code> now sourced from <code>project_members</code> instead of <code>time_entries</code>

5. Test Execution

Run Unit Tests

```
npm run test           # single run
npm run test:watch     # watch mode
npm run test:coverage  # with coverage report
```

Run UAT / E2E Tests

Prerequisites:

1. Ensure `.env.local` contains `NEXT_PUBLIC_SUPABASE_URL` , `NEXT_PUBLIC_SUPABASE_ANON_KEY` , `TEST_USER_EMAIL` , `TEST_USER_PASSWORD`
2. Start the dev server: `npm run dev`
3. In a second terminal:

```
npx playwright test           # headless
npx playwright test --reporter=list # with per-test output
npx playwright test --ui       # interactive UI mode
npx playwright test --debug    # step-through debugger
```

6. Coverage Summary

Area	Unit Tests	UAT Tests
Duration formatting	✓ (9 cases)	—
Currency formatting	✓ (6 cases)	—
Earnings calculation (level rates)	✓ (11 cases)	—
Invoice hours classification	✓ (9 cases)	✓ (hours summary panel)
Invoice line generation	✓ (8 cases)	✓ (generate flow)
Cashflow KPI calculation	✓ (7 cases)	✓ (analytics section)
PM authorization logic	✓ (8 cases)	✓ (timesheets review tab)
Timesheet status transitions	✓ (9 cases)	✓ (submit/review flow)
Authentication	—	✓
Dashboard / Navigation	—	✓
Timer start/stop	—	✓
Project management	—	✓
Invoice history	—	✓

Total: 67 unit tests · 40 UAT tests = 107 automated tests